Group 2

BUILDING MOBILE APPLICATION FOR APPOINTMENT SCHEDULING AND RECORDS TRACKING SYSTEM - THU DUC REGIONAL GENERAL HOSPITAL

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Information System Analysis and Design Course

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AGENDA OVERVIEW

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01 Introduction to the Hospital

02 Identifying the Business Problem & Current Issues Business Impact

03 Project Requirements

04 The Solution - Our Mobile App Mobile App Features and Functions

05 Demo app

06 Anticipated Impact and Conclusion

1. INTRODUCTION TO THE HOSPITAL



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1.INTRODUCTION TO THE HOSPITAL



Established:

1978, initially took over a monastery in Tan Phu commune, Thu Duc district (now Linh Trung ward).

Current Scale:

- Over 2,000 outpatient visits daily, 700+ inpatients.
- 8 functional rooms, 25 clinical and subclinical departments.
- Staff: 600+ members, including experienced and dedicated doctors.

1.INTRODUCTION TO THE HOSPITAL



New Hospital Project:

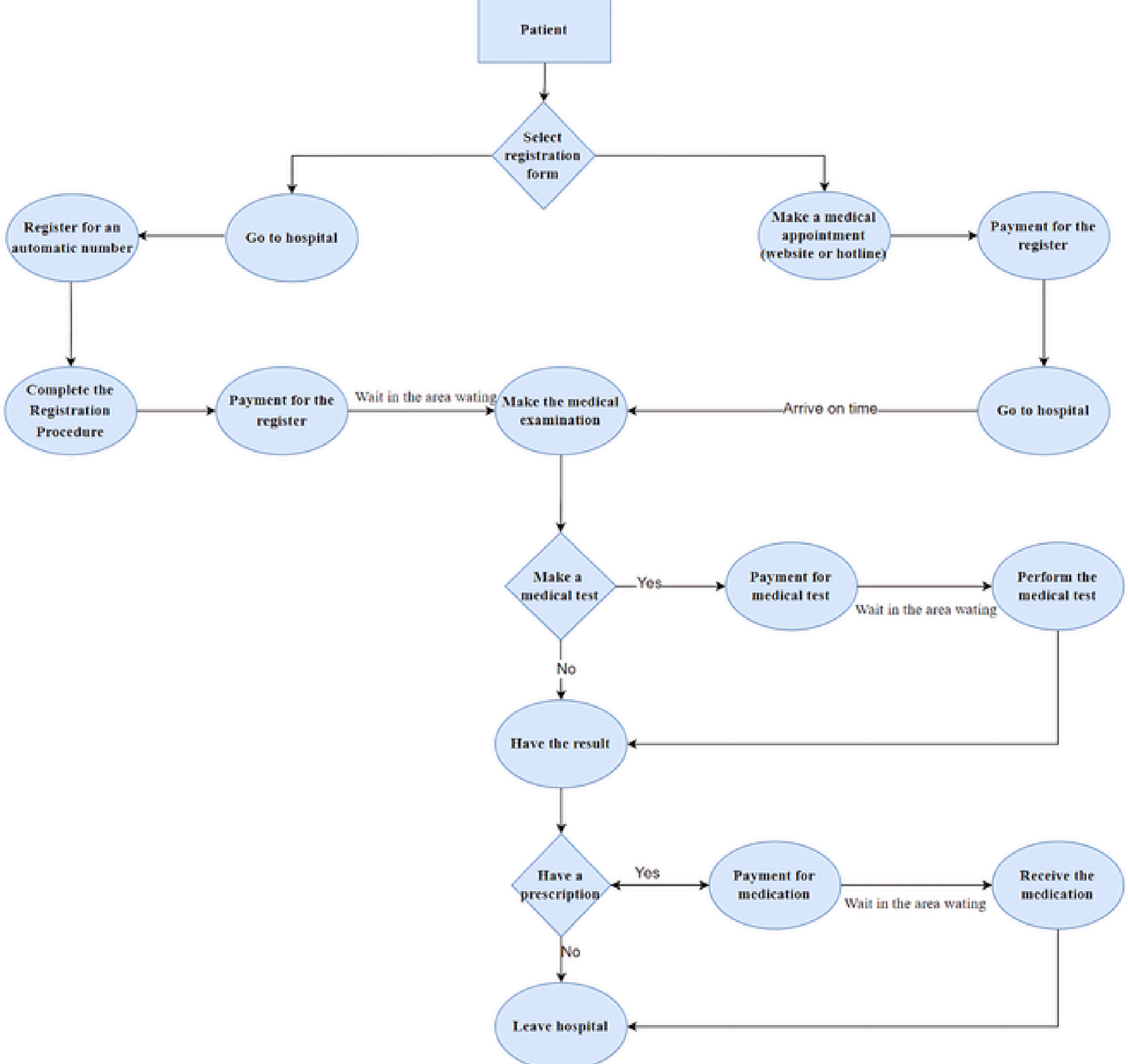
- Approved by Ho Chi Minh City People's Committee in 2019, with an investment of VND 1,915 billion.
- Construction began in late 2021; completion expected by **April 2025**.
- Design: 1 basement, 10 floors, 58,000m^2, 1,000-bed capacity, 500 high-quality inpatient beds.
- **Purpose**: To ease **inner-city hospital overcrowding** and improve healthcare for Thu Duc and nearby areas.

02. BUSINESS PROBLEM & CURRENT ISSUES BUSINESS IMPACT



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CURRENT SYSTEM



BUSINESS PROBLEM





Patient reviews highlighting long wait times and administrative delays (Source: Google).

BUSINESS PROBLEM

Key Issues

- Inefficient Appointment Scheduling
- High Administrative Workload
- Disjointed Data Management
- Absence of Real-Time Updates
- Resource Misallocation
- Limited Patient Access to Medical Records

03. PROJECT REQUIREMENTS



FACT FINDINGS

Observation

- **Direct Observation:** Analysts observe patient flow, staff workflows, and identify inefficiencies in patient registration and scheduling at Thu Duc Regional General Hospital.
- **Environmental Observation:** Assessment of the hospital's physical layout and signage to understand patient navigation and accessibility of information.

Document Review

- **Analyzing Existing System Documentation:** Analysts examined Thu Duc hospital's current appointment system documents.
- **Reviewing Hospital Policies:** Reviewed policies on appointment scheduling, cancellations, rescheduling, and assessed service level agreements or targets related to patient wait times.

Interviews

- **Patient Interviews:** Insights from diverse patient groups to understand their experiences, challenges, and expectations for a new appointment system.
- **Staff Interviews:** Insights from staff about current system challenges and needs for a more efficient workflow.

Research

- **Benchmarking:** Evaluation of best practices in hospital appointment systems.
- **Market Research:** Understanding trends in patient expectations and technology use in healthcare for system development.

AS-IS PROCESS
-> PROPOSED SYSTEM

As-is process

- Manual, paper-based processes
- Long wait times
- Limited online functionality
- Lack of integration with hospital systems

Proposed objectives

- Convenient Appointment Scheduling
- User-Friendly Interface
- Secure Online Payments
- System Integration
- Streamlined Check-In Process

PROJECT REQUIREMENTS

Functional Requirements

- User registration, profile management, and secure scheduling.
- Real-time medical record access.
- Queue management, check-in notifications, and secure payments.
- Sync with hospital systems.

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Non-functional Requirements

- Security: Data protection and encryption.
- Scalability: Supports high user volumes.
- Reliability: High availability with minimal downtime.

Business Constraints

- Compliance with healthcare regulations (decree 13/2023/NĐ-CP).
- Integration with existing hospital systems.
- Compatibility with iOS and Android.

04. THE SOLUTION - OUR MOBILE APP MOBILE APPLICATION FEATURES AND FUNCTIONS



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FUNCTIONS & FEATUERS

Login/Signup

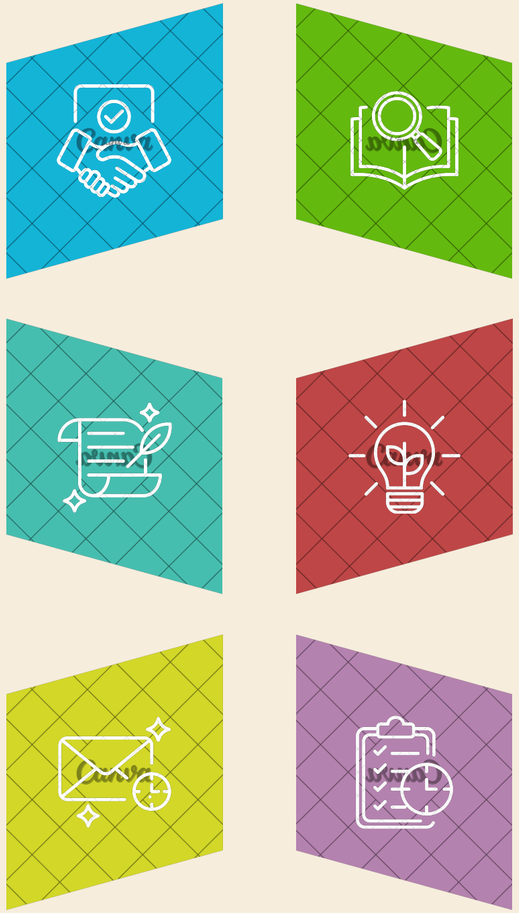
Allows users to create a new account or log into an existing one using their phone number. An OTP code is sent for verification, simplifying access and enhancing security.

Manage User Account

Enables users to update personal information, such as contact details and payment methods, ensuring profiles are current and manageable.

Booking

Facilitates appointment scheduling by letting users select a preferred doctor, department, and time slot, improving access to healthcare services.



Payment

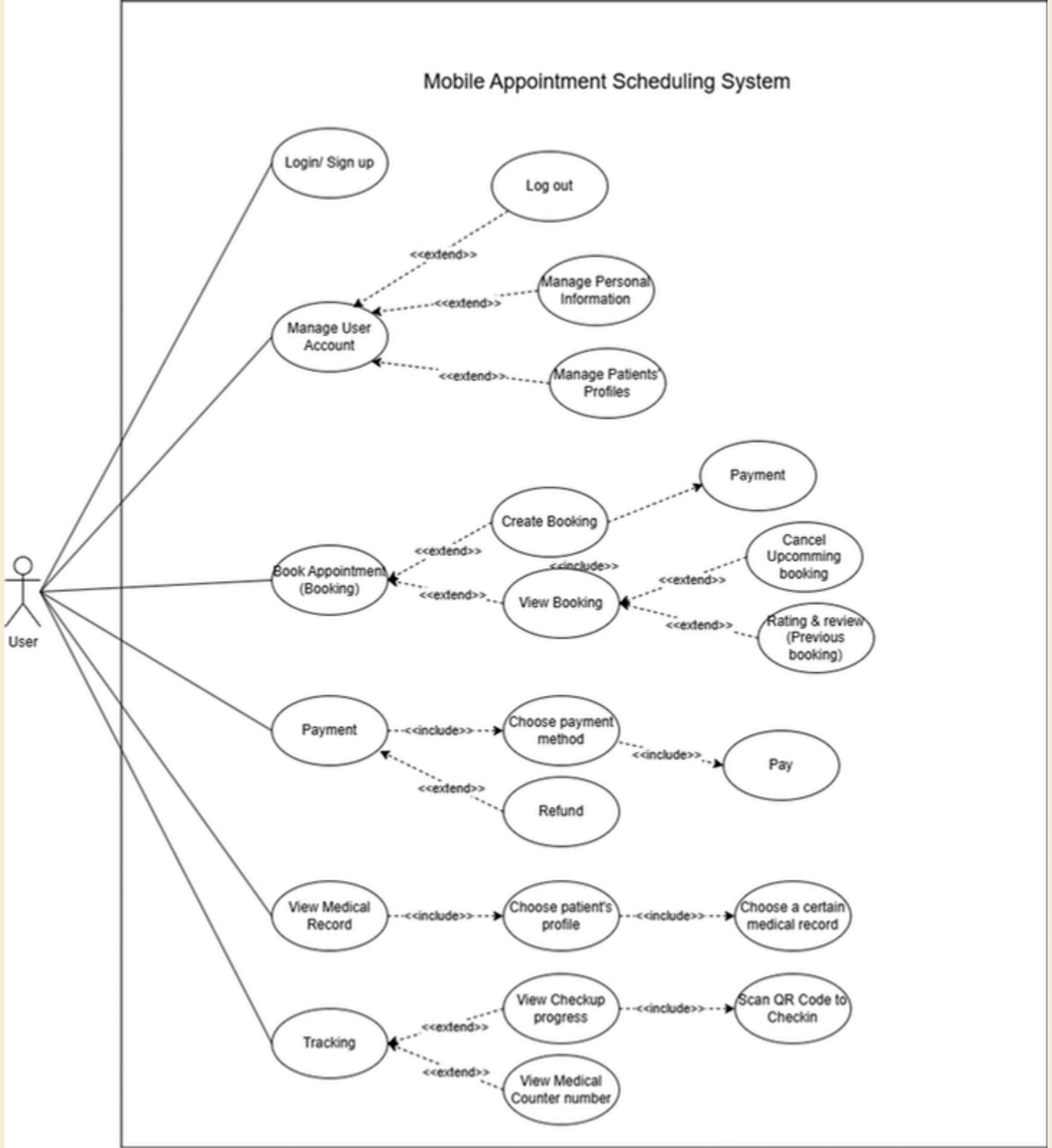
Integrates secure payment processing for appointments, supporting both direct payments and insurance claims for a seamless transaction experience.

View medical Record

Allows patients to access their medical history, diagnoses, prescriptions, and test results, providing transparency and easy follow-up management.

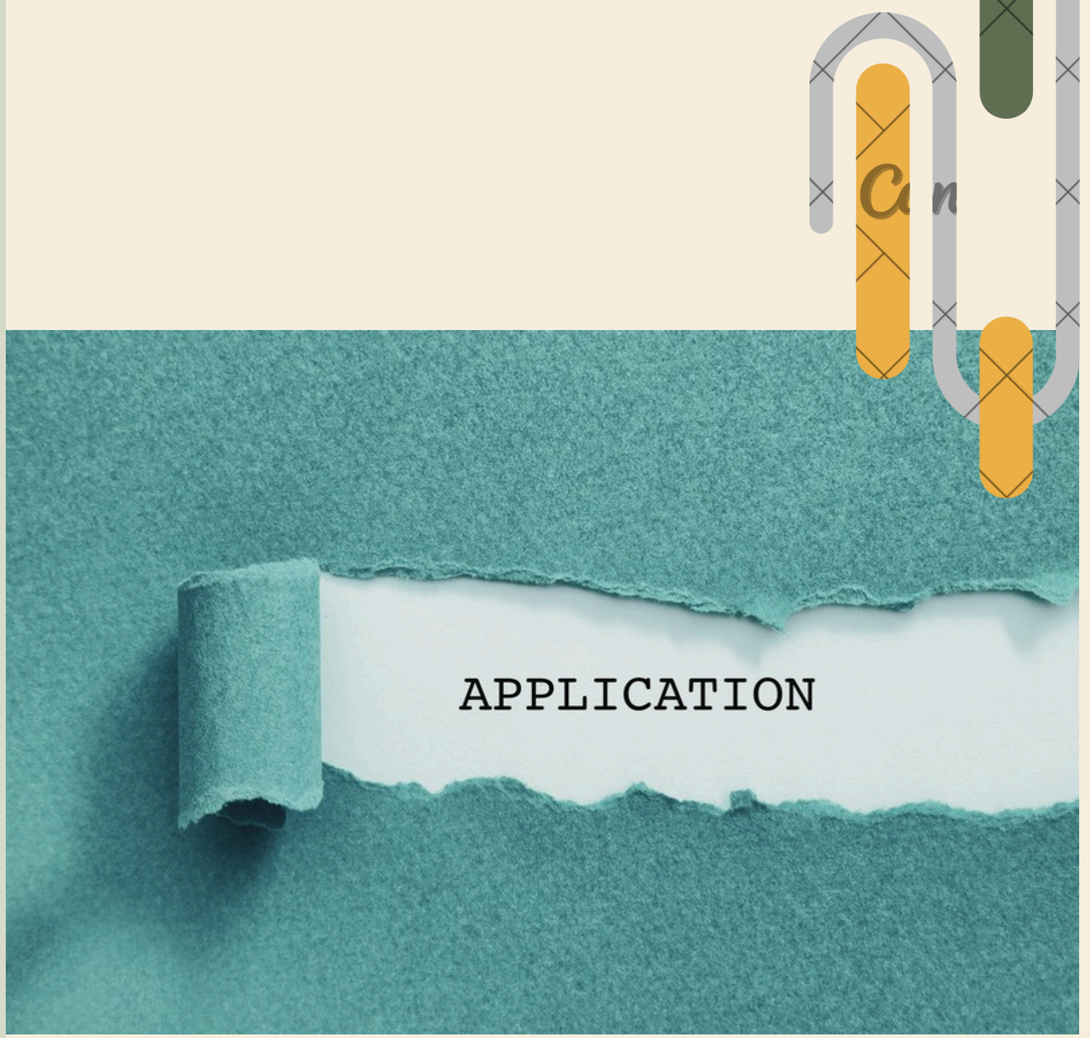
Tracking

Sends real-time notifications and updates on appointment statuses, test results, and other important reminders, keeping patients informed throughout their healthcare journey



USE CASE DIAGRAM OVERAL APPLICATION

05. DEMO APPLICATION



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06. ANTICIPATED IMPACT AND CONCLUSION



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CONCLUSION



Anticipated Impact

The mobile scheduling system will elevate patient experience through reduced wait times, real-time updates, and streamlined record access.



Conclusion

This digital transformation boosts operational efficiency and strengthens the hospital’s reputation for high-quality care, positioning it as a leading healthcare provider prepared to meet future demands.

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THANK YOU

12 May, 2024